

# Status

## Boba — Feature Status (all components)

**Revision:** 5 **Last modified:** 2026-06-16T12:15:00Z **Scope:** Every system component, service, infrastructure piece, and client app of the Boba project — one row per REAL unit (endpoint / handler / client method / component control / plugin / subcommand / script), grouped by component. **Authority:** assembled by READ-ONLY repo inventory (codegraph + grep + source reading) on 2026-06-15, expanded to per-unit granularity (§11.4.118 discovery-pressure) on 2026-06-16. Cross-references AGENTS.md, CLAUDE.md, docs/REMAINING\_WORK\_PLAN.md.

Captured-evidence-driven (§11.4.5 / §11.4.45 / §11.4.86). Every feature row cites a real source file (file:line where load-bearing), endpoint, command, or control. **No invented features** (§11.4.6). As of Rev 5 the **Video** column is DEFINITIVE for every row — there is no bare PENDING. Each cell is exactly one of: VIDEO-CONFIRMED — <file> (the feature is shown on-screen in a committed recording), PENDING (UI — film next) (a user-visible control/dialog not yet *individually* filmed), or N/A (no UI — test-covered + exercised by <journey>) (a non-user-visible unit — REST endpoint / handler / client method / parser / scanner / service / crypto / repo / script — which has no screen of its own and is confirmed by its cited tests plus the UI/CLI journey that drives it). **Honest classification (§11.4.6/§11.4.143): a row is VIDEO-CONFIRMED only when a real recording actually shows it.**

Per-row Video tally (288 feature rows): **25 VIDEO-CONFIRMED, 17 PENDING (UI — film next), 246 N/A (no UI — test-covered).**

Column legend: - **Impl** = implemented / partial / stub / missing - **Wired** = reachable on a running default-profile stack (yes / no / opt-in / host) - **Tests** = test types present + cite file (unit / integration / e2e / security / stress-chaos / property / contract); none if no covering test found - **Validation** = honest validation posture (tested-green-in-suite / unit-only / live-evidence-captured / not-validated / operator-blocked) - **Video** = DEFINITIVE video-recording status — one of VIDEO-CONFIRMED — <file> / PENDING (UI — film next) / N/A (no UI — test-covered + exercised by <journey>)

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# Video-recording confirmations (§11.4.143 / §11.4.83)

Real-use recordings (Claude-vision analyzed — the HelixAgent ensemble vision is a proven stub, see docs/qa/recording-readiness-20260615/) in /Volumes/T7/Downloads/Recordings/ (project-prefix boba-). Per-row Video cells stay PENDING where a feature has not been *individually* filmed; the recordings below confirm the headline user-facing flows on-screen.

| Recording                                               | Surface                                                  | Features confirmed                                                                                                                                                                                                                                                                                           | Verdict                                         |
|---------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| boba-cli-orchestrator-demo.mp4                          | CLI (boba-ctl)                                           | status / health / list — 4 services running & healthy                                                                                                                                                                                                                                                        | docs/qa/recordings-20260615/boba-cli-verdict.md |
| boba-web-search-flow.mp4                                | Web dashboard                                            | launch → search “debian” → live results → “829 results (288 merged)” complete dashboard load (29 trackers, qBit connected), result rows render <b>qBit</b> + <b>Download</b> buttons (clickable), tab nav (Results/ Active Downloads/ Trackers/ Schedules/ Hooks), Jackett / jackett credentials page, theme | boba-web-search-flow-verdict.md                 |
| boba-web-feature-tour.mp4 + boba-web-dashboard-tour.mp4 | Web dashboard                                            | content-script <b>detects a magnet + appends a</b>                                                                                                                                                                       | boba-web-feature-tour-verdict.md                |
| boba-extension-{scan-detect,popup,options,tour}.mp4     | BobaLink extension (MV3, real --load-extension Chromium) |                                                                                                                                                                                                                                                                                                              | boba-extension-verdict.md                       |

| Recording | Surface | Features confirmed                                                                                                                                                                    | Verdict |
|-----------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|           |         | <b>MAGNET badge</b><br>(non-torrent correctly skipped);<br>popup UI (header, server status, Refresh + Send All, Detected-torrents);<br>options page (7 settings tabs + server config) |         |

**Honest scope (§11.4.6, no bluff):** - **Backend services** (download-proxy / merge / boba-jackett / Go) verified **200 server-side inside the VM** and exercised through the web UI recordings. - **Console errors** seen in the web recordings are an **SSH-tunnel SSE/poll artifact** (macOS VM), NOT product defects — every endpoint is 200 in the VM. See the feature-tour verdict. - **BobaLink extension:**  **VIDEO-CONFIRMED** — the prior --load-extension tooling gap was closed by building extension/scripts/record-features.mjs (loads the built MV3 artifact into a real Chromium, records the scan/popup/options journeys). Plus its passing test suite. - **TUI / mobile / desktop:** do not exist in this project (server-side + Angular web + browser extension only) — nothing to record.

## Component summary

| # | Component                                              | Service / Port                                               | Features cataloged |
|---|--------------------------------------------------------|--------------------------------------------------------------|--------------------|
| 1 | Download Proxy + Merge Search Service (Python/FastAPI) | qbittorrent-proxy :7186 / :7187                              | 68                 |
| 2 | qBitTorrent-go backend (Go/Gin)                        | qbittorrent-proxy-go :7186/:7187/:7188 (opt-in --profile go) | 47                 |
| 3 | boba-jackett (Go)                                      | boba-jackett :7189                                           | 26                 |
| 4 | Tracker plugins (plugins/*.py)                         | run inside qbittorrent-proxy                                 | 30                 |

| # | Component                                          | Service / Port     | Features cataloged |
|---|----------------------------------------------------|--------------------|--------------------|
| 5 | Angular 21<br>frontend<br>dashboard<br>(frontend/) | served from :7187  | 34                 |
| 6 | BobaLink<br>browser<br>extension<br>(extension/)   | WXT MV3 client     | 39                 |
| 7 | WebUI bridge<br>(webui-<br>bridge.py)              | host process :7188 | 4                  |
| 8 | Infrastructure /<br>CLI / shell<br>scripts         | host               | 40                 |

**Total features cataloged: 288**

## 1. Download Proxy + Merge Search Service (Python/ FastAPI) — :7186 / :7187

Entry: `download-proxy/src/main.py` (starts legacy proxy thread + `uvicorn api:app`). This is the **shipped default product**.

### 1a. Merge Search Service — REST API (`download-proxy/src/api/routes.py`)

| Feature                                                                                                                | Impl        | Wired | Tests                                                                                                                                                          | Validation                                        |
|------------------------------------------------------------------------------------------------------------------------|-------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| POST <code>/api/v1/search</code> — async fan-out search, returns <code>search_id</code> ( <code>routes.py:283</code> ) | implemented | yes   | integration tests/ <code>integration/test_merge_api.py</code> ; e2e <code>test_full_pipeline.py</code> ; stress <code>test_merge_search_stress_chaos.py</code> | tested-green-in-suite + LIVE                      |
| POST <code>/api/v1/search/sync</code> — synchronous search ( <code>routes.py:377</code> )                              | implemented | yes   | integration <code>test_merge_api.py</code> ; stress <code>test_search_stress.py</code>                                                                         | tested-green-in-suite; resets over tunnel (RW-08) |
| GET <code>/api/v1/search/stream/{id}</code> — SSE result                                                               | implemented | yes   | integration <code>test_realtime_streaming.py</code> , <code>test_streaming_browser.py</code>                                                                   | tested-green-in-suite                             |

| Feature                                                                                              | Impl        | Wired | Tests                                                                                                     | Validation                                          |
|------------------------------------------------------------------------------------------------------|-------------|-------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| stream<br>(routes.py:547)                                                                            |             |       |                                                                                                           |                                                     |
| GET /api/v1/<br>search/{id} —<br>poll search state<br>(routes.py:586)                                | implemented | yes   | integration test_merge_api.py                                                                             | tested-<br>green-in-<br>suite                       |
| POST /api/v1/<br>search/{id}/<br>abort — cancel<br>in-flight search<br>(routes.py:619)               | implemented | yes   | integration test_merge_api.py                                                                             | tested-<br>green-in-<br>suite                       |
| GET /api/v1/<br>downloads/<br>active — qBit<br>active torrents<br>(routes.py:629)                    | implemented | yes   | integration test_buttons_api.py                                                                           | tested-<br>green-in-<br>suite                       |
| POST /api/v1/<br>auth/<br>qbittorrent —<br>qBit WebUI login<br>(routes.py:671)                       | implemented | yes   | integration test_login_actions.py                                                                         | tested-<br>green-in-<br>suite                       |
| POST /api/v1/<br>download — add<br>torrent to qBit<br>(magnet/url)<br>(routes.py:958)                | implemented | yes   | integration<br>test_button_functions.py,<br>test_buttons_api.py; security<br>test_ssrf_and_magnet_auth.py | tested-<br>green-in-<br>suite;<br>SSRF gap<br>RW-03 |
| POST /api/v1/<br>download/upload<br>— upload .torrent<br>file to qBit<br>(routes.py:1123)            | implemented | yes   | unit tests/unit/api_layer/<br>test_download_upload.py                                                     | unit-only<br>+<br>integration                       |
| POST /api/v1/<br>download/file<br>— fetch tracker<br>.torrent w/ auth<br>cookies<br>(routes.py:1212) | implemented | yes   | integration test_buttons_api.py;<br>security<br>test_ssrf_and_magnet_auth.py                              | tested-<br>green-in-<br>suite;<br>SSRF gap<br>RW-03 |
| POST /api/v1/<br>magnet — resolve<br>result → magnet<br>link<br>(routes.py:1279)                     | implemented | yes   | integration test_magnet_dialog.py,<br>test_buttons_api.py                                                 | tested-<br>green-in-<br>suite; auth<br>gap<br>RW-04 |
| GET /api/v1/<br>theme — read<br>theme state<br>(routes.py:61)                                        | implemented | yes   | contract<br>test_crossapp_theme_contract.py                                                               | tested-<br>green-in-<br>suite                       |
| PUT /api/v1/<br>theme — persist                                                                      | implemented | yes   | e2e test_theme_runtime.py,<br>test_crossapp_theme.py                                                      | tested-<br>green-in-<br>suite                       |

| Feature                                                                                               | Impl        | Wired | Tests                      | Validation                    |
|-------------------------------------------------------------------------------------------------------|-------------|-------|----------------------------|-------------------------------|
| theme<br>(routes.py:67)<br>GET /api/v1/<br>theme/stream —<br>SSE theme<br>broadcast<br>(routes.py:76) | implemented | yes   | e2e test_crossapp_theme.py | tested-<br>green-in-<br>suite |

## 1b. Auth API — per-tracker (download-proxy/src/api/auth.py)

| Feature                                                                                              | Impl        | Wired | Tests                                                                              | Validation                                                   | Vi                                    |
|------------------------------------------------------------------------------------------------------|-------------|-------|------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------|
| GET /auth/<br>rutracker/<br>status —<br>RuTracker<br>session state<br>(auth.py:56)                   | implemented | yes   | integration<br>test_auth_and_links.py; e2e<br>test_live_stack_evidence.py          | partial —<br>CAPTCHA<br>operator-<br>blocked<br>(BOB-008)    | N/<br>—<br>co<br>ex<br>we<br>Tr<br>au |
| GET /auth/<br>rutracker/<br>captcha —<br>fetch<br>CAPTCHA<br>image<br>(auth.py:108)                  | implemented | yes   | integration test_auth_and_links.py                                                 | partial —<br>operator-<br>blocked<br>(BOB-008)               | N/<br>—<br>co<br>ex<br>we<br>Tr<br>au |
| POST /auth/<br>rutracker/<br>login —<br>username/<br>password +<br>CAPTCHA<br>login<br>(auth.py:214) | implemented | yes   | integration test_auth_and_links.py                                                 | partial —<br>operator-<br>blocked<br>(BOB-008)               | N/<br>—<br>co<br>ex<br>we<br>Tr<br>au |
| POST /auth/<br>rutracker/<br>cookie-login<br>— cookie-<br>based login<br>(auth.py:298)               | implemented | yes   | integration test_auth_and_links.py                                                 | partial —<br>live-gated                                      | N/<br>—<br>co<br>ex<br>we<br>Tr<br>au |
| GET /auth/<br>nmclub/<br>status —<br>NNM-Club<br>session state<br>(auth.py:350)                      | implemented | yes   | e2e<br>test_live_stack_evidence.py:265<br>(skips — SOURCE→ARTIFACT drift<br>RW-07) | partial —<br>route 404 on<br>running<br>container<br>(RW-07) | N/<br>—<br>co<br>ex<br>we<br>Tr<br>au |
| POST /auth/<br>nmclub/                                                                               | implemented | yes   | e2e test_live_stack_evidence.py                                                    | partial —<br>live-gated                                      | N/<br>—                               |

| Feature                                                                               | Impl           | Wired | Tests                                          | Validation                                                                  | Vi                               |
|---------------------------------------------------------------------------------------|----------------|-------|------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------|
| login —<br>NNM-Club<br>login<br>(auth.py:407)                                         |                |       |                                                |                                                                             | co<br>ex<br>we<br>Tr<br>au<br>N/ |
| GET /auth/<br>status — all-<br>tracker<br>aggregate<br>session state<br>(auth.py:479) | implemented    | yes   | integration test_auth_state_ui.py              | tested-green-<br>in-suite                                                   | co<br>ex<br>we<br>Tr<br>au       |
| POST /auth/<br>qbittorrent/<br>logout —<br>qBit WebUI<br>logout<br>(auth.py:532)      | implemented    | yes   | integration test_login_actions.py              | tested-green-<br>in-suite                                                   | N/<br>—<br>co<br>ex<br>we<br>log |
| Kinozal auth<br>(status/login)<br>REST route                                          | <b>missing</b> | no    | none — no kinozal/login route in<br>auth.py    | not-<br>implemented<br>(search path<br>uses cookies<br>via<br>orchestrator) | N/<br>im<br>—                    |
| IPTorrents auth<br>(status/login)<br>REST route                                       | <b>missing</b> | no    | none — no iptorrents/login route<br>in auth.py | not-<br>implemented<br>(env creds +<br>cookie flow<br>via Jackett<br>UI)    | N/<br>im<br>—                    |

### 1c. Search orchestration internals (download-proxy/src/ merge\_service/search.py)

| Feature                                                            | Impl        | Wired | Tests                              |
|--------------------------------------------------------------------|-------------|-------|------------------------------------|
| Concurrency caps<br>MAX_CONCURRENT_SEARCHES (429 on<br>saturation) | implemented | yes   | concurrency test_orchestrator_     |
| Per-search tracker semaphore<br>MAX_CONCURRENT_TRACKERS            | implemented | yes   | concurrency test_tracker_semaphore |

| Feature                                                                       | Impl        | Wired | Tests                                                                         |
|-------------------------------------------------------------------------------|-------------|-------|-------------------------------------------------------------------------------|
| Public-tracker subprocess fan-out<br>(NDJSON via patched novaprinter)         | implemented | yes   | e2e<br>test_public_trackers_return_<br>stress<br>test_search_orchestration_st |
| Public-tracker deadline kill<br>PUBLIC_TRACKER_DEADLINE_SECONDS               | implemented | yes   | chaos test_tracker_failure.py                                                 |
| Private-tracker aiohttp+cookie path<br>(rutracker/kinozal/nnmclub/iptorrents) | implemented | yes   | integration test_tracker_auth_<br>gated)                                      |
| Per-tracker TrackerSearchStat real-<br>time stats                             | implemented | yes   | contract test_tracker_stats_con<br>property test_tracker_stats_pro            |
| Stderr error classification<br>(_classify_plugin_stderr)                      | implemented | yes   | chaos test_tracker_failure.py                                                 |



| Feature                                           | Impl        | Wired | Tests                                   |
|---------------------------------------------------|-------------|-------|-----------------------------------------|
| TTL-bounded data stores (cachetools)<br>— no leak | implemented | yes   | memory<br>test_orchestrator_caches_boun |

### 1d. Deduplication (download-proxy/src/merge\_service/deduplicator.py)

| Feature                                 | Impl        | Wired | Tests                                                                                        | Validation            | Video                                                                        |
|-----------------------------------------|-------------|-------|----------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------|
| Tier 1 dedup — metadata (imdb/tmdb) key | implemented | yes   | property<br>test_deduplicator_properties.py;<br>benchmark<br>test_deduplication_benchmark.py | tested-green-in-suite | N/A (no UI — test-covered + exercised by web search flow “288 merged result) |
| Tier 2 dedup — infohash key             | implemented | yes   | property<br>test_deduplicator_properties.py                                                  | tested-green-in-suite | N/A (no UI — test-covered + exercised by web search flow “288 merged result) |
| Tier 3 dedup — name+size key            | implemented | yes   | property<br>test_deduplicator_properties.py                                                  | tested-green-in-suite | N/A (no UI — test-covered + exercised by web search flow “288 merged result) |

| Feature                                     | Impl        | Wired | Tests                                                                                        | Validation            | Video                                                                       |
|---------------------------------------------|-------------|-------|----------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------|
| Tier 4 dedup — fuzzy Levenshtein name match | implemented | yes   | property<br>test_deduplicator_properties.py;<br>benchmark<br>test_deduplication_benchmark.py | tested-green-in-suite | N/A (no UI — test-covered + exercise by web search flow “288 merged result) |

### 1e. Enrichment — per metadata source (download-proxy/src/merge\_service/enricher.py)

| Feature                            | Impl        | Wired | Tests                                   | Validation                                |
|------------------------------------|-------------|-------|-----------------------------------------|-------------------------------------------|
| Enrichment: OMDb (enricher.py:113) | implemented | yes   | stress<br>test_enricher_stress_chaos.py | tested-green-in-suite; needs OMDb_API_KEY |
| Enrichment: TMDB                   | implemented | yes   | stress<br>test_enricher_stress_chaos.py | tested-green; needs TMDB_API_KEY          |
| Enrichment: TVMaze                 | implemented | yes   | stress<br>test_enricher_stress_chaos.py | tested-green-in-suite                     |

| Feature                 | Impl        | Wired | Tests                                   | Validation            |
|-------------------------|-------------|-------|-----------------------------------------|-----------------------|
| Enrichment: AniList     | implemented | yes   | stress<br>test_enricher_stress_chaos.py | tested-green-in-suite |
| Enrichment: MusicBrainz | implemented | yes   | stress<br>test_enricher_stress_chaos.py | tested-green-in-suite |
| Enrichment: OpenLibrary | implemented | yes   | stress<br>test_enricher_stress_chaos.py | tested-green-in-suite |

#### 1f. Tracker validation (download-proxy/src/merge\_service/validator.py)

| Feature                                                        | Impl        | Wired | Tests                                         | Validation                                     | Video                                                    |
|----------------------------------------------------------------|-------------|-------|-----------------------------------------------|------------------------------------------------|----------------------------------------------------------|
| Tracker validation<br>HTTP scrape BEP 48<br>(validator.py:101) | implemented | yes   | none<br>dedicated<br>found<br>(covered<br>via | not-<br>validated<br>(no<br>dedicated<br>test) | N/A (no<br>UI —<br>exercised<br>by web<br>search<br>flow |

| Feature                                                       | Impl        | Wired | Tests                                             | Validation                                     | Video                                                                                                                          |
|---------------------------------------------------------------|-------------|-------|---------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Tracker validation<br>UDP scrape BEP 15<br>(validator.py:162) | implemented | yes   | merge<br>tests)<br><br>none<br>dedicated<br>found | not-<br>validated<br>(no<br>dedicated<br>test) | seeds/<br>leechers<br>column)<br><br>N/A (no<br>UI —<br>exercised<br>by web<br>search<br>flow<br>seeds/<br>leechers<br>column) |

### 1g. Hooks (download-proxy/src/api/hooks.py + merge\_service/hooks.py)

| Feature                                                           | Impl        | Wired | Tests                                                                                                       | Val                              |
|-------------------------------------------------------------------|-------------|-------|-------------------------------------------------------------------------------------------------------------|----------------------------------|
| GET /api/v1/hooks — list<br>hooks (hooks.py:105)                  | implemented | yes   | unit test_hooks_coverage.py;<br>concurrency<br>test_hooks_race.py; security<br>test_hooks_schedules_auth.py | test<br>gre<br>suit<br>gap<br>RW |
| POST /api/v1/hooks — create<br>hook (hooks.py:111)                | implemented | yes   | unit test_hooks_coverage.py;<br>concurrency<br>test_hooks_race.py                                           | test<br>gre<br>suit<br>gap<br>RW |
| DELETE /api/v1/hooks/{id}<br>— delete hook (hooks.py:167)         | implemented | yes   | unit test_hooks_coverage.py                                                                                 | test<br>gre<br>suit<br>gap<br>RW |
| GET /api/v1/hooks/logs —<br>hook execution logs<br>(hooks.py:179) | implemented | yes   | unit test_hooks_coverage.py                                                                                 | test<br>gre<br>suit              |

| Feature                                                                         | Impl        | Wired | Tests                                  | Val                             |
|---------------------------------------------------------------------------------|-------------|-------|----------------------------------------|---------------------------------|
| Hook event dispatch (8 events: search_start..validation_complete) (hooks.py:32) | implemented | yes   | concurrency<br>test_hooks_race.py      | test<br>gre<br>suit             |
| Hook subprocess execution + sandbox (merge_service/hooks.py)                    | implemented | yes   | security<br>test_no_shell_injection.py | test<br>gre<br>san<br>har<br>RW |

## 1h. Scheduler (download-proxy/src/api/scheduler.py + merge\_service/scheduler.py)

| Feature                                                           | Impl        | Wired | Tests                                    | Validation                                          | Vi                                                                                                                                              |
|-------------------------------------------------------------------|-------------|-------|------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| GET /api/v1/schedules — list schedules (scheduler.py:39)          | implemented | yes   | security<br>test_hooks_schedules_auth.py | tested-<br>green-in-<br>suite; auth<br>gap<br>RW-02 | N.<br>U<br>te<br>co<br>ex<br>by<br>da<br>Sc<br>ta<br>N.<br>U<br>te<br>co<br>ex<br>by<br>da<br>Sc<br>ta<br>N.<br>U<br>te<br>co<br>ex<br>by<br>da |
| POST /api/v1/schedules — create schedule (scheduler.py:64)        | implemented | yes   | security<br>test_hooks_schedules_auth.py | tested-<br>green-in-<br>suite; auth<br>gap<br>RW-02 | N.<br>U<br>te<br>co<br>ex<br>by<br>da<br>Sc<br>ta<br>N.<br>U<br>te<br>co<br>ex<br>by<br>da                                                      |
| GET /api/v1/schedules/{id} — get schedule (scheduler.py:87)       | implemented | yes   | security<br>test_hooks_schedules_auth.py | tested-<br>green-in-<br>suite                       | N.<br>U<br>te<br>co<br>ex<br>by<br>da<br>Sc<br>ta<br>N.<br>U<br>te<br>co<br>ex<br>by<br>da                                                      |
| PATCH /api/v1/schedules/{id} — update schedule (scheduler.py:108) | implemented | yes   | security<br>test_hooks_schedules_auth.py | tested-<br>green-in-<br>suite                       | N.<br>U<br>te<br>co<br>ex<br>by<br>da                                                                                                           |

| Feature                                                               | Impl        | Wired | Tests                                       | Validation            | Vi                                                                                                          |
|-----------------------------------------------------------------------|-------------|-------|---------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------|
| DELETE /api/v1/schedules/{id} — delete schedule (scheduler.py:126)    | implemented | yes   | security<br>test_hooks_schedules_auth.py    | tested-green-in-suite | Sc<br>tal<br>N/<br>U/<br>te<br>co<br>ex<br>by<br>da<br>Sc<br>tal<br>N/<br>U/<br>te<br>co<br>in<br>tic<br>dr |
| Scheduler engine (load/save/tick driver) (merge_service/scheduler.py) | implemented | yes   | integration<br>test_dashboard_automation.py | tested-green-in-suite |                                                                                                             |

## 1i. App-level routes / health / config / dashboard (download-proxy/src/api/\_\_init\_\_.py)

| Feature                                     | Impl        | Wired | Tests                                       | Validation            | Vi                                      |
|---------------------------------------------|-------------|-------|---------------------------------------------|-----------------------|-----------------------------------------|
| GET /health (__init__.py:168)               | implemented | yes   | integration<br>test_live_containers.py      | tested-green-in-suite | N/A<br>test<br>exe<br>bob<br>CL         |
| GET /api/v1/bridge/health (__init__.py:173) | implemented | yes   | integration<br>test_bridge_root_liveness.py | tested-green-in-suite | N/A<br>test<br>exe<br>wel<br>hea<br>ind |
| GET /api/v1/config (__init__.py:247)        | implemented | yes   | integration test_merge_api.py               | tested-green-in-suite | N/A<br>test<br>exe<br>wel<br>boo        |
| GET /api/v1/stats (__init__.py:276)         | implemented | yes   | observability<br>test_metrics_exist.py      | tested-green-in-suite | N/A<br>test<br>exe<br>wel<br>stat       |
| Jinja2 dashboard GET / (__init__.py:337)    | implemented | yes   | integration<br>test_dashboard_rendering.py  | tested-green-in-suite | N/A<br>ser<br>das<br>SPA<br>das         |

| Feature                                                 | Impl        | Wired | Tests                                      | Validation                    | View                                                                                                                                                                                                                 |
|---------------------------------------------------------|-------------|-------|--------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jinja2 dashboard<br>GET /dashboard<br>(__init__.py:342) | implemented | yes   | integration<br>test_dashboard_rendering.py | tested-<br>green-in-<br>suite | itself<br>VIII<br>CO<br>boba<br>featu<br>tough<br>N/A<br>serv<br>das<br>SPA<br>das<br>itself<br>VIII<br>CO<br>boba<br>featu<br>tough<br>N/A<br>serv<br>dee<br>incl<br>jackett<br>VIII<br>CO<br>via<br>featu<br>tough |
| SPA catch-all GET /<br>{path:path}<br>(__init__.py:347) | implemented | yes   | integration<br>test_dashboard_rendering.py | tested-<br>green-in-<br>suite |                                                                                                                                                                                                                      |

### 1j. Supporting subsystems (Python)

| Feature                                                                                                           | Impl        | Wired | Tests                                                                                                                                   | View |
|-------------------------------------------------------------------------------------------------------------------|-------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------|------|
| Retry helpers<br>(merge_service/<br>retry.py)                                                                     | implemented | yes   | covered indirectly                                                                                                                      |      |
| Jackett autoconfig (Python,<br>merge_service/<br>jackett_autoconfig.py)<br>— legacy; canonical is<br>boba-jackett | implemented | yes   | contract<br>test_jackett_autoconfig_contract.py;<br>e2e test_jackett_autoconfig_e2e.py;<br>benchmark<br>test_jackett_autoconfig_perf.py |      |
| Credential scrubbing log<br>filter (config/<br>log_filter.py)                                                     | implemented | yes   | security test_credential_scrubbing.py                                                                                                   |      |

| Feature | Impl | Wired | Tests |  |
|---------|------|-------|-------|--|
|---------|------|-------|-------|--|

## 1k. Legacy download proxy (plugins/download\_proxy.py :7186)

| Feature                                                                                          | Impl        | Wired | Tests                                         | Validation                    |
|--------------------------------------------------------------------------------------------------|-------------|-------|-----------------------------------------------|-------------------------------|
| HTTP proxy →<br>qBittorrent WebUI w/<br>tracker URL<br>interception<br>(DownloadHandler<br>:767) | implemented | yes   | integration<br>test_bridge_header_rewrite.py  | tested-<br>green-in-<br>suite |
| Theme injection into<br>proxied WebUI<br>(plugins/<br>theme_injector.py)                         | implemented | yes   | integration<br>test_bridge_theme_injection.py | tested-<br>green-in-<br>suite |

## 2. qBitTorrent-go backend (Go/Gin) — opt-in -- profile go

Entry: qBitTorrent-go/cmd/qbittorrent-proxy/main.go (routes registered :54-101). **Skeleton/rewrite-in-progress** (per AGENTS.md + docs/migration/PARITY\_GAPS.md). NOT running by default.

### 2a. HTTP handlers (internal/api/\*.go)

| Feature                                                         | Impl        | Wired  | Tests                       | Validation             |
|-----------------------------------------------------------------|-------------|--------|-----------------------------|------------------------|
| GET /health —<br>HealthHandler (api/<br>health.go:5)            | implemented | opt-in | go internal/api/api_test.go | unit-only<br>(go test) |
| GET /api/v1/config —<br>ConfigHandler (api/<br>download.go:210) | implemented | opt-in | go coverage_test.go         | unit-only              |



| Feature                                                                          | Impl                                                                           | Wired  | Tests                                                | Validation                       |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------|------------------------------------------------------|----------------------------------|
| GET /api/v1/bridge/health —<br>BridgeHealthHandler<br>(api/download.go:189)      | implemented                                                                    | opt-in | go coverage_test.go                                  | unit-only                        |
| POST /api/v1/search —<br>SearchHandler (api/search.go:14)                        | implemented<br>(proxies<br>qBit search;<br>no plugin<br>fan-out)               | opt-in | go api_test.go,<br>coverage_test.go                  | unit-only                        |
| POST /api/v1/search/sync —<br>SearchSyncHandler<br>(api/search.go:50)            | implemented                                                                    | opt-in | go coverage_test.go                                  | unit-only                        |
| GET /api/v1/search/stream/:id —<br>SearchStreamHandler<br>SSE (api/search.go:91) | implemented                                                                    | opt-in | go coverage_test.go                                  | unit-only                        |
| GET /api/v1/search/:id —<br>GetSearchHandler<br>(api/search.go:144)              | implemented                                                                    | opt-in | go<br>getsearch_results_test.go,<br>coverage_test.go | unit-only                        |
| POST /api/v1/search/:id/abort —<br>AbortSearchHandler<br>(api/search.go:177)     | implemented                                                                    | opt-in | go coverage_test.go                                  | unit-only                        |
| POST /api/v1/download —<br>DownloadHandler<br>(api/download.go:17)               | <b>partial —<br/>returns<br/>mock<br/>success, no<br/>qBit<br/>integration</b> | opt-in | go coverage_test.go                                  | unit-only;<br>functional<br>hole |
| POST /api/v1/download/file —<br>DownloadFileHandler<br>(api/download.go:41)      | implemented                                                                    | opt-in | go coverage_test.go                                  | unit-only                        |

| Feature                                                                      | Impl                                             | Wired  | Tests                                        | Validation                 |
|------------------------------------------------------------------------------|--------------------------------------------------|--------|----------------------------------------------|----------------------------|
| POST /api/v1/magnet — MagnetHandler (api/download.go:86)                     | implemented                                      | opt-in | go api_test.go, magnet_stress_chaos_test.go  | unit + stress-chaos (go)   |
| GET /api/v1/downloads/active — ActiveDownloadsHandler (api/download.go:147)  | <b>stub — returns empty array, no qBit query</b> | opt-in | go coverage_test.go                          | unit-only; functional hole |
| POST /api/v1/auth/qbittorrent — QBittorrentAuthHandler (api/download.go:156) | implemented                                      | opt-in | go coverage_test.go                          | unit-only                  |
| GET /api/v1/theme — GetThemeHandler (api/theme.go:90)                        | implemented                                      | opt-in | go theme_hardening_test.go, coverage_test.go | unit-only                  |
| PUT /api/v1/theme — PutThemeHandler (api/theme.go:96)                        | implemented                                      | opt-in | go theme_hardening_test.go                   | unit-only                  |
| GET /api/v1/hooks — ListHooksHandler (api/hooks.go:91)                       | implemented (store only, no dispatch)            | opt-in | go api_test.go, coverage_test.go             | unit-only                  |
| POST /api/v1/hooks — CreateHookHandler (api/hooks.go:97)                     | implemented (store only)                         | opt-in | go api_test.go                               | unit-only                  |
| DELETE /api/v1/hooks/:id — DeleteHookHandler (api/hooks.go:109)              | implemented (store only)                         | opt-in | go api_test.go                               | unit-only                  |

| Feature                                                                                       | Impl                        | Wired  | Tests               | Validation                                                                        |
|-----------------------------------------------------------------------------------------------|-----------------------------|--------|---------------------|-----------------------------------------------------------------------------------|
| GET /api/v1/schedules<br>—<br>ListSchedulesHandler<br>(api/<br>scheduler_api.go:86)           | implemented<br>(store only) | opt-in | go coverage_test.go | unit-only;<br><b>no driver<br/>loop<br/>(RW-10),<br/>schedules<br/>never fire</b> |
| POST /api/v1/schedules<br>—<br>CreateScheduleHandler<br>(api/<br>scheduler_api.go:92)         | implemented<br>(store only) | opt-in | go coverage_test.go | unit-only;<br>never fires<br>(RW-10)                                              |
| DELETE /api/v1/<br>schedules/:id —<br>DeleteScheduleHandler<br>(api/<br>scheduler_api.go:104) | implemented<br>(store only) | opt-in | go coverage_test.go | unit-only                                                                         |

## 2b. qBittorrent Web API client (internal/client/\*.go)

| Feature                                                                       | Impl        | Wired  | Tests                        | Validation | Video                                                                             |
|-------------------------------------------------------------------------------|-------------|--------|------------------------------|------------|-----------------------------------------------------------------------------------|
| NewClient —<br>construct + login<br>(client/<br>client.go:18)                 | implemented | opt-in | go client/<br>client_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| Login — POST /<br>api/v2/auth/<br>login, store SID<br>(client/<br>auth.go:11) | implemented | opt-in | go client/<br>auth_test.go   | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| IsAuthenticated<br>— SID presence                                             | implemented | opt-in | go client/<br>auth_test.go   | unit-only  | N/A (no<br>UI —<br>go-test-                                                       |

| Feature                                                                                       | Impl        | Wired  | Tests                          | Validation | Video                                                                             |
|-----------------------------------------------------------------------------------------------|-------------|--------|--------------------------------|------------|-----------------------------------------------------------------------------------|
| (client/<br>auth.go:43)                                                                       |             |        |                                |            | covered<br>client<br>method;<br>opt-in<br>profile)                                |
| GetSID — thread-<br>safe SID accessor<br>(client/<br>client.go:39)                            | implemented | opt-in | go client/<br>client_test.go   | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| GetTorrents —<br>GET /api/v2/<br>torrents/info<br>(client/<br>torrents.go:13)                 | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| AddTorrent —<br>POST /api/v2/<br>torrents/add<br>URL (client/<br>torrents.go:30)              | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| AddTorrentFile<br>— POST multipart<br>/api/v2/<br>torrents/add<br>(client/<br>torrents.go:54) | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| GetAppVersion —<br>GET /api/v2/<br>app/version<br>(client/<br>torrents.go:87)                 | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| StartSearch —<br>POST /api/v2/<br>search/start<br>(client/<br>search.go:27)                   | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;                       |

| Feature                                                                           | Impl        | Wired  | Tests                          | Validation | Video                                                                                                   |
|-----------------------------------------------------------------------------------|-------------|--------|--------------------------------|------------|---------------------------------------------------------------------------------------------------------|
| GetSearchResults<br>— GET /api/v2/<br>search/results<br>(client/<br>search.go:55) | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | opt-in<br>profile)<br>N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile) |
| StopSearch —<br>POST /api/v2/<br>search/stop<br>(client/<br>search.go:84)         | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile)                       |
| SearchStatus —<br>GET /api/v2/<br>search/status<br>(client/<br>search.go:100)     | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile)                       |
| ListPlugins —<br>GET /api/v2/<br>search/plugins<br>(client/<br>search.go:121)     | implemented | opt-in | go client/<br>coverage_test.go | unit-only  | N/A (no<br>UI —<br>go-test-<br>covered<br>client<br>method;<br>opt-in<br>profile)                       |

## 2c. Services (internal/service/\*.go)

| Feature                                                                                              | Impl                                                                       | Wired  | Tests                                                  |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------|--------------------------------------------------------|
| Merge search orchestrator<br>RunSearch poll/dedup loop<br>(service/<br>merge_search.go:183)          | implemented<br>(qBit search<br>only; no<br>python-style<br>plugin fan-out) | opt-in | go service/service_test.go,<br>runsearch_dedup_test.go |
| StartSearch /<br>GetSearchStatus /<br>AbortSearch lifecycle<br>(service/<br>merge_search.go:101–143) | implemented                                                                | opt-in | go<br>getsearchstatus_hardening_test.go                |
|                                                                                                      | implemented                                                                | opt-in | go runsearch_dedup_test.go                             |

| Feature                                                               | Impl                                                                  | Wired  | Tests               |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------|--------|---------------------|
| Dedup key<br>(FileURL→name+size)<br>(service/<br>merge_search.go:269) |                                                                       |        |                     |
| FetchTorrent tracker fetch<br>(service/<br>merge_search.go:276)       | <b>stub —<br/>returns “not<br/>yet<br/>implemented”</b>               | opt-in | none                |
| Stats counters (service/<br>merge_search.go:280)                      | implemented                                                           | opt-in | go coverage_test.go |
| SSE broker pub/sub<br>(service/sse_broker.go)                         | implemented<br>(defined but<br>not wired —<br>handlers SSE<br>inline) | opt-in | none dedicated      |
| Metadata enricher (Go)                                                | <b>missing<br/>(RW-11)</b>                                            | no     | none                |
| Scheduler driver loop (Go)                                            | <b>missing<br/>(RW-10) —<br/>CRUD only,<br/>no cron<br/>execution</b> | no     | none                |

## 2d. webui-bridge binary (Go) + cross-cutting

| Feature                                                       | Impl                               | Wired  | Tests                                | Validation        | Video                                                                    |
|---------------------------------------------------------------|------------------------------------|--------|--------------------------------------|-------------------|--------------------------------------------------------------------------|
| webui-bridge<br>binary (cmd/<br>webui-<br>bridge/<br>main.go) | partial<br>(health +<br>root only) | opt-in | none dedicated                       | not-<br>validated | N/A (no UI<br>of its own<br>— proxy<br>binary; opt-<br>in profile)       |
| CORS +<br>logging<br>middleware<br>(internal/<br>middleware/) | implemented                        | opt-in | go middleware/<br>middleware_test.go | unit-only         | N/A (no UI<br>— go-test-<br>covered<br>middleware;<br>opt-in<br>profile) |
| Models / data<br>types<br>(internal/<br>models/)              | implemented                        | opt-in | go models/<br>models_test.go         | unit-only         | N/A (no UI<br>— go-test-<br>covered<br>data types;<br>opt-in<br>profile) |

| Feature                                     | Impl        | Wired  | Tests                       | Validation | Video                                                       |
|---------------------------------------------|-------------|--------|-----------------------------|------------|-------------------------------------------------------------|
| Env config loader (internal/config/)        | implemented | opt-in | go config/config_test.go    | unit-only  | N/A (no UI — go-test-covered config loader; opt-in profile) |
| Log redactor (internal/logging/redactor.go) | implemented | opt-in | go logging/redactor_test.go | unit-only  | N/A (no UI — go-test-covered redactor; opt-in profile)      |

### 3. boba-jackett (Go) — :7189

Entry: qBittorrent-go/cmd/boba-jackett/main.go; router internal/jackettapi/router.go. Owns Jackett credentials + indexer overrides + autoconfig run history; encrypted SQLite at /config/boba.db.

#### 3a. HTTP endpoints (internal/jackettapi/router.go + handlers)

| Feature                                                                         | Impl        | Wired | Tests                                                                    | Validation              | Video                                                 |
|---------------------------------------------------------------------------------|-------------|-------|--------------------------------------------------------------------------|-------------------------|-------------------------------------------------------|
| GET /healthz (router.go:60, health.go)                                          | implemented | yes   | go jackettapi/health_test.go                                             | unit (go)               | N/A — covered endpoints                               |
| GET /openapi.json (router.go:64, openapi.go)                                    | implemented | yes   | go jackettapi/openapi_test.go; contract tests/contract/openapi_test.go   | unit + contract (go)    | N/A — covered endpoints                               |
| GET /api/v1/jackett/credentials — list (router.go:67, credentials.go)           | implemented | yes   | go credentials_test.go; integration tests/integration/jackett_db_test.go | unit + integration (go) | N/A — covered endpoints, example website, create page |
| POST /api/v1/jackett/credentials — create/update (router.go:67, credentials.go) | implemented | yes   | go credentials_test.go                                                   | unit + integration (go) | N/A — covered endpoints, example website, create page |

| Feature                                                                     | Impl        | Wired | Tests                                                                | Validation              | View                                    |
|-----------------------------------------------------------------------------|-------------|-------|----------------------------------------------------------------------|-------------------------|-----------------------------------------|
| DELETE /api/v1/jackett/credentials/{name} (router.go:77, credentials.go)    | implemented | yes   | go credentials_test.go                                               | unit + integration (go) | N/A — coverage excluded we create pages |
| GET /api/v1/jackett/indexers — list configured (router.go:86, indexers.go)  | implemented | yes   | go jackettapi/indexers_test.go                                       | unit (go)               | N/A — coverage excluded we index pages  |
| POST/PATCH/DELETE /api/v1/jackett/indexers/{id} (router.go:93, indexers.go) | implemented | yes   | go indexers_test.go                                                  | unit (go)               | N/A — coverage excluded we index pages  |
| POST /api/v1/jackett/indexers/{id}/test — test config (indexers.go:239)     | implemented | yes   | go indexers_test.go                                                  | unit (go)               | N/A — coverage excluded we index pages  |
| GET /api/v1/jackett/catalog — indexer catalog (router.go:112, catalog.go)   | implemented | yes   | go jackettapi/catalog_test.go; benchmark catalog_bench_test.go       | unit + bench (go)       | N/A — coverage excluded we catalog      |
| POST /api/v1/jackett/catalog/refresh (router.go:119, catalog.go)            | implemented | yes   | go catalog_test.go                                                   | unit (go)               | N/A — coverage excluded we catalog      |
| GET /api/v1/jackett/autoconfig/runs — run history (router.go:137, runs.go)  | implemented | yes   | go jackettapi/runs_test.go; e2e tests/e2e/jackett_management_test.go | unit + e2e (go)         | N/A — coverage excluded we History      |
| GET /api/v1/jackett/autoconfig/runs/{id} — run detail                       | implemented | yes   | go runs_test.go                                                      | unit (go)               | N/A — coverage excluded                 |



| Feature                                                                                                | Impl        | Wired | Tests                                                     | Validation      | Video                                          |
|--------------------------------------------------------------------------------------------------------|-------------|-------|-----------------------------------------------------------|-----------------|------------------------------------------------|
| (router.go:144, runs.go)<br>POST /api/v1/jackett/autocfg/run — trigger run<br>(router.go:151, runs.go) | implemented | yes   | go runs_test.go; e2e tests/e2e/jackett_management_test.go | unit + e2e (go) | we<br>His<br>N/<br>—<br>cov<br>ex<br>we<br>His |
| GET/POST /api/v1/jackett/overrides<br>(router.go:160, overrides.go)                                    | implemented | yes   | go jackettapi/overrides_test.go                           | unit (go)       | N/<br>—<br>cov<br>enc                          |
| DELETE /api/v1/jackett/overrides/{env}<br>(router.go:170, overrides.go)                                | implemented | yes   | go overrides_test.go                                      | unit (go)       | N/<br>—<br>cov<br>enc                          |
| Admin auth middleware (admin/admin on mutating routes) (jackettapi/auth_middleware.go)                 | implemented | yes   | go auth_middleware_test.go                                | unit (go)       | N/<br>—<br>cov<br>mi                           |
| Hardened CORS middleware (jackettapi/cors_middleware.go)                                               | implemented | yes   | go cors_middleware_test.go                                | unit (go)       | N/<br>—<br>cov<br>mi                           |

### 3b. Autoconfig engine + persistence (internal/jackett/, internal/db/)

| Feature                                     | Impl        | Wired | Tests                                             | Validation        | Video                                                                        |
|---------------------------------------------|-------------|-------|---------------------------------------------------|-------------------|------------------------------------------------------------------------------|
| Jackett autocfg engine (jackett/autocfg.go) | implemented | yes   | go jackett/autocfg_test.go, autocfg_bench_test.go | unit + bench (go) | N/A (no UI — go-test-covered engine; exercised by web / jackett autocfg run) |
| Fuzzy indexer matcher (jackett/matcher.go)  | implemented | yes   | go jackett/matcher_test.go                        | unit (go)         | N/A (no UI — go-test-covered matcher)                                        |
| Jackett HTTP client (jackett/client.go)     | implemented | yes   | go jackett/client_test.go                         | unit (go)         | N/A (no UI — go-test-covered client)                                         |

| Feature                                              | Impl        | Wired | Tests                                                                 | Validation           | Video                                               |
|------------------------------------------------------|-------------|-------|-----------------------------------------------------------------------|----------------------|-----------------------------------------------------|
| Encrypted SQLite — AES-256-GCM crypto (db/crypto.go) | implemented | yes   | go db/crypto_test.go; security tests/security/credential_leak_test.go | unit + security (go) | N/A (no UI — go-test + security-covered crypto)     |
| SQLite migrations (db/migrate.go)                    | implemented | yes   | go db/migrate_test.go                                                 | unit (go)            | N/A (no UI — go-test-covered migrations)            |
| Credential / indexer / runs repos (db/repos/*.go)    | implemented | yes   | go db/repos/*_test.go                                                 | unit (go)            | N/A (no UI — go-test-covered repos)                 |
| .env bootstrap import (bootstrap/*.go)               | implemented | yes   | go bootstrap/bootstrap_test.go                                        | unit (go)            | N/A (no UI — go-test-covered bootstrap)             |
| Master-key ensure (first-boot) (bootstrap/*.go)      | implemented | yes   | go bootstrap/bootstrap_test.go                                        | unit (go)            | N/A (no UI — go-test-covered first-boot key ensure) |
| .env file parse/write (envfile/*.go)                 | implemented | yes   | go envfile/parse_test.go, write_test.go                               | unit (go)            | N/A (no UI — go-test-covered env file I/O)          |

## 4. Tracker plugins (plugins/\*.py)

Plugin contract: class with url, name, supported\_categories, search(), download\_torrent(). Installed into container by install-plugin.sh. The install-plugin.sh PLUGINS array names **44 plugins** (curated install set); the plugins/ tree carries **27 \*.py files** (24 tracker/aggregator plugins + 3 support: nova2, novaprinter, socks, helpers, env\_loader, theme\_injector, download\_proxy). Many curated names have NO matching file in this repo tree (the install script fetches/expects them elsewhere) — flagged honestly below.

### 4a. Plugins present in plugins/ AND named in the curated array

| Plugin  | Type   | Impl        | Wired (curated?) | Tests                               |
|---------|--------|-------------|------------------|-------------------------------------|
| eztv.py | public | implemented | yes (curated)    | stress test_plugin_parsers_stress_c |

| Plugin           | Type   | Impl        | Wired<br>(curated?) | Tests                                  |
|------------------|--------|-------------|---------------------|----------------------------------------|
| limetorrents.py  | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| piratebay.py     | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| solidtorrents.py | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| torlock.py       | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |

| Plugin           | Type                  | Impl        | Wired<br>(curated?) | Tests                                  |
|------------------|-----------------------|-------------|---------------------|----------------------------------------|
| rutor.py         | public<br>(anonymous) | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| nyaa.py          | public                | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| anilibra.py      | public                | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| bitsearch.py     | public                | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| gamestorrents.py | public                | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |

| Plugin           | Type   | Impl        | Wired<br>(curated?) | Tests                                  |
|------------------|--------|-------------|---------------------|----------------------------------------|
| kickass.py       | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| megapeer.py      | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| tokyotoshokan.py | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |
| torrentgalaxy.py | public | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c |

| Plugin          | Type                            | Impl        | Wired<br>(curated?) | Tests                                                                                         |
|-----------------|---------------------------------|-------------|---------------------|-----------------------------------------------------------------------------------------------|
| torrentkitty.py | public                          | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c                                                        |
| yts.py          | public                          | implemented | yes<br>(curated)    | stress<br>test_plugin_parsers_stress_c                                                        |
| jackett.py      | aggregator                      | implemented | yes<br>(curated)    | covered via jackett autoconfig tests                                                          |
| iptorrents.py   | private<br>(freeleech-<br>only) | implemented | yes<br>(curated)    | integration tests/integration/<br>test_iptorrents.py                                          |
| rutracker.py    | private<br>(CAPTCHA)            | implemented | yes<br>(curated)    | integration test_tracker_auth_t<br>stress<br>test_plugin_parsers_stress_c<br>(ReDoS §11.4.85) |

| Plugin     | Type                 | Impl        | Wired<br>(curated?) | Tests                           |
|------------|----------------------|-------------|---------------------|---------------------------------|
| kinozal.py | private              | implemented | yes<br>(curated)    | integration test_tracker_auth_l |
| nnmclub.py | private<br>(cookies) | implemented | yes<br>(curated)    | integration test_tracker_auth_l |

#### 4b. Plugin support modules

| Module                                                 | Type    | Impl        | Wired | Tests                  | Validation                    | Vi                                                     |
|--------------------------------------------------------|---------|-------------|-------|------------------------|-------------------------------|--------------------------------------------------------|
| nova2.py — nova3<br>search harness                     | support | implemented | yes   | covered via search e2e | tested-<br>green-in-<br>suite | N/<br>UI<br>co<br>sea<br>har<br>ex<br>by<br>sea<br>flo |
| novaprinter.py —<br>result emitter<br>(NDJSON-patched) | support | implemented | yes   | covered via search e2e | tested-<br>green-in-<br>suite | N/<br>UI<br>co<br>res<br>em<br>ex                      |

| Module                                                 | Type    | Impl        | Wired | Tests                                           | Validation            | View                                                                                                                                                                                                                                                           |
|--------------------------------------------------------|---------|-------------|-------|-------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| helpers.py — HTTP/retrieve helpers                     | support | implemented | yes   | covered via search e2e                          | tested-green-in-suite | by search flow<br>N/UI<br>covered by helpers.exe by search flow<br>N/UI<br>integration SC support covered including<br>N/UI<br>covered by create loader.exe by Tracker tab<br>N/UI<br>covered by resource Do button<br>N/UI<br>covered by resource Do (from po |
| socks.py — SOCKS proxy support                         | support | implemented | yes   | covered indirectly                              | not-validated         |                                                                                                                                                                                                                                                                |
| env_loader.py — env/credential loader for plugins      | support | implemented | yes   | covered via tracker auth tests                  | tested-green-in-suite |                                                                                                                                                                                                                                                                |
| Per-plugin download_torrent() path (.torrent fetch)    | feature | implemented | yes   | integration test_buttons_api.py (download/file) | tested-green-in-suite |                                                                                                                                                                                                                                                                |
| Freeleech-only download policy (IPTorrents [free] tag) | policy  | implemented | yes   | integration test_iporrents.py                   | tested-green-in-suite |                                                                                                                                                                                                                                                                |



#### 4c. Discrepancy rows (honest, §11.4.6)

| Plugin                                                                                                                                                                                                                                                                                                                                                           | Type                         | Impl                                                                                                            | Wired (curated?)      | Tests | Validation                                                                            | Video                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------|-------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| torrentproject /<br>torrentscsv                                                                                                                                                                                                                                                                                                                                  | public<br>(curated<br>names) | named in<br>install-<br>plugin.sh<br>PLUGINS<br>array but<br>NO<br>matching<br>plugins/<br>*.py file<br>in tree | curated-<br>name-only | none  | not-<br>implemented-<br>in-repo-tree<br>(install script<br>expects them<br>elsewhere) | N/A (not<br>implemented<br>in repo tree<br>— curated<br>name, no<br>plugin file)   |
| 23 other curated<br>names absent from<br>tree:<br>academictorrents,<br>ali213,<br>audiobookbay,<br>bitru, bt4g,<br>btsow,<br>extratorrent,<br>glotorrents,<br>linuxtracker,<br>one337x,<br>pctorrent,<br>pirateiro,<br>rockbox, snowfl,<br>therarbg,<br>torrentdownload,<br>torrentfunk,<br>xfsb, yihua,<br>yourbittorrent (+<br>torrentproject,<br>torrentscsv) | public<br>(curated<br>names) | named in<br>curated<br>array; NO<br>plugins/<br>*.py in<br>this tree                                            | curated-<br>name-only | none  | not-<br>implemented-<br>in-repo-tree<br>(name in<br>install array;<br>file absent)    | N/A (not<br>implemented<br>in repo tree<br>— curated<br>names, no<br>plugin files) |

NOTE (§11.4.6, CORRECTION vs Rev 2): the install-plugin.sh PLUGINS array contains **44 entries**, not 12. Of those, ~21 have a matching plugins/\*.py file in this repo; the rest are curated names with no file present in the tree. Stated as fact, not asserted as working.

## 5. Angular 21 frontend dashboard (frontend/)

Standalone SPA, signals-based; built to download-proxy/src/ui/dist/frontend, served from :7187. Vitest unit tests; Playwright e2e under frontend/e2e.

## 5a. Dashboard component controls (`components/dashboard/dashboard.component.ts`)

| Feature                                                                               | Impl        | Wired | Tests                                                                                                                                                         |
|---------------------------------------------------------------------------------------|-------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Search box (query signal) + submit                                                    | implemented | yes   | vitest<br><code>dashboard.component.spec.ts</code> ; py<br>integration<br><code>test_dashboard_rendering.py</code> ,<br><code>test_ui_comprehensive.py</code> |
| Result grid — sortable (name/type/size/seeds/leechers/quality/sources)                | implemented | yes   | vitest<br><code>dashboard.component.spec.ts</code>                                                                                                            |
| Results tab                                                                           | implemented | yes   | py <code>test_ui_comprehensive.py</code>                                                                                                                      |
| Active Downloads tab + view ( <code>dashboard.component.ts:745 loadDownloads</code> ) | implemented | yes   | py <code>test_buttons_api.py</code>                                                                                                                           |
| Trackers tab + per-tracker auth chips (expandable)                                    | implemented | yes   | py <code>test_auth_state_ui.py</code>                                                                                                                         |
| Schedules tab                                                                         | implemented | yes   | py<br><code>test_dashboard_automation.py</code>                                                                                                               |
| Hooks tab                                                                             | implemented | yes   |                                                                                                                                                               |

| Feature                                       | Impl        | Wired | Tests                                                                   |
|-----------------------------------------------|-------------|-------|-------------------------------------------------------------------------|
|                                               |             |       | py<br>test_dashboard_automation.py                                      |
| Per-row Magnet button → magnet dialog trigger | implemented | yes   | vitest magnet-<br>dialog.component.spec.ts; py<br>test_magnet_dialog.py |
| Per-row add-to-qBittorrent (qBit) button      | implemented | yes   | py test_button_functions.py,<br>test_buttons_api.py                     |
| Per-row Download (.torrent) button            | implemented | yes   | py test_buttons_api.py                                                  |
| Per-row processing spinner (isBusy())         | implemented | yes   | vitest<br>dashboard.component.spec.ts                                   |
| Bridge health indicator + retry button        | implemented | yes   | py<br>test_bridge_root_liveness.py                                      |
| Tracker-stat dialog trigger                   | implemented | yes   | py contract<br>test_tracker_stats_contract.py                           |

## 5b. Dialog + UI components

| Feature                                                                              | Impl        | Wired | Tests                                                                                                 | Valida                     |
|--------------------------------------------------------------------------------------|-------------|-------|-------------------------------------------------------------------------------------------------------|----------------------------|
| Magnet dialog — readonly textarea + Copy/Open/Add/Close (magnet-dialog.component.ts) | implemented | yes   | vitest magnet-<br>dialog.component.spec.ts; py<br>test_magnet_dialog.py                               | tested.<br>green-<br>suite |
| qBit login dialog — user/pass/remember/Login/Cancel (qbit-login-dialog.component.ts) | implemented | yes   | vitest qbit-login-<br>dialog.component.spec.ts; py<br>test_login_actions.py,<br>test_auth_state_ui.py | tested.<br>green-<br>suite |
| Theme picker — palette dropdown + light/dark toggle (theme-picker.component.ts)      | implemented | yes   | vitest theme-<br>picker.component.spec.ts; py<br>e2e test_theme_runtime.py,<br>test_crossapp_theme.py | tested.<br>green-<br>suite |

| Feature                                                                                                            | Impl        | Wired | Tests                                                                                    | Validation         |
|--------------------------------------------------------------------------------------------------------------------|-------------|-------|------------------------------------------------------------------------------------------|--------------------|
| Confirm dialog — title/message/Confirm/Cancel/backdrop (confirm-dialog.component.ts)                               | implemented | yes   | vitest confirm-dialog.component.spec.ts                                                  | unit (vitest)      |
| Tracker-stat dialog — JSON view + Copy + status badge + notes table + Esc-close (tracker-stat-dialog.component.ts) | implemented | yes   | vitest tracker-stat-dialog.component.spec.ts; py contract test_tracker_stats_contract.py | tested-green-suite |
| Toast container — info/success/warning/error auto-dismiss (toast-container.component.ts)                           | implemented | yes   | vitest toast-container.component.spec.ts                                                 | unit (vitest)      |
| Site footer (site-footer.component.ts)                                                                             | implemented | yes   | vitest                                                                                   | unit (vitest)      |

## 5c. Services (frontend/src/app/services/)

| Feature                                                                                                                                        | Impl        | Wired | Tests                                                                                            | Validation            | Vi                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|--------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------|
| API service — HTTP wrapper (search/download/magnet/auth/schedules/hooks/downloads/stats/config), 15s timeout, runtime baseUrl (api.service.ts) | implemented | yes   | vitest api.service.spec.ts                                                                       | unit (vitest)         | N/<br>—<br>ser<br>ex<br>ev<br>da<br>cal         |
| SSE service — EventSource live result streaming + bearer-token (sse.service.ts)                                                                | implemented | yes   | vitest sse.service.spec.ts; py integration test_realtime_streaming.py, test_streaming_browser.py | tested-green-in-suite | N/<br>—<br>ser<br>ex<br>we<br>flo<br>res<br>str |
| Theme service — palette/mode + localStorage + cross-app sync (theme.service.ts)                                                                | implemented | yes   | vitest theme.service.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py               | tested-green-in-suite | N/<br>—<br>ser<br>ex<br>we<br>pic               |
| Toast service — toast queue state (toast.service.ts)                                                                                           | implemented | yes   | vitest toast.service.spec.ts                                                                     | unit (vitest)         | N/<br>—<br>ser<br>ex<br>we<br>no                |

| Feature                                                                                  | Impl        | Wired | Tests                                         | Validation       | Video                                       |
|------------------------------------------------------------------------------------------|-------------|-------|-----------------------------------------------|------------------|---------------------------------------------|
| Dialog service —<br>promise-based confirm<br>state<br>( <code>dialog.service.ts</code> ) | implemented | yes   | vitest<br><code>dialog.service.spec.ts</code> | unit<br>(vitest) | N/A<br>—<br>series<br>examples<br>we<br>dia |

## 5d. Jackett page (`frontend/src/app/jackett/`)

| Feature                                                                                                                                                        | Impl        | Wired | Tests                                                           | Validation       | Video |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------|-----------------------------------------------------------------|------------------|-------|
| Credentials — list + Add/Edit/<br>Delete (per-row,<br><code>window.confirm</code> ) + banners<br>( <code>jackett/credentials/credentials.component.ts</code> ) | implemented | yes   | vitest<br><code>credentials.component.spec.ts</code>            | unit<br>(vitest) |       |
| Credential edit dialog — name/<br>username/password/cookies/<br>Save/Cancel ( <code>credential-edit-dialog.component.ts</code> )                               | implemented | yes   | vitest<br><code>credential-edit-dialog.component.spec.ts</code> | unit<br>(vitest) |       |
| Indexers — Configured/<br>Catalog/History tab container<br>( <code>jackett/indexers/indexers.component.ts</code> )                                             | implemented | yes   | vitest<br><code>indexers.component.spec.ts</code>               | unit<br>(vitest) |       |
| Indexers — Configured tab<br>(list + per-row remove)                                                                                                           | implemented | yes   | vitest<br><code>indexers.component.spec.ts</code>               | unit<br>(vitest) |       |
| Indexers — Catalog tab (list +<br>per-row add)                                                                                                                 | implemented | yes   | vitest<br><code>indexers.component.spec.ts</code>               | unit<br>(vitest) |       |
| Indexers — History tab<br>(autoconfig run history)                                                                                                             | implemented | yes   | vitest<br><code>indexers.component.spec.ts</code>               | unit<br>(vitest) |       |
| Indexer add dialog —<br>credential dropdown + Save/<br>Cancel + error banner<br>( <code>indexer-add-dialog.component.ts</code> )                               | implemented | yes   | vitest<br><code>indexer-add-dialog.component.spec.ts</code>     | unit<br>(vitest) |       |
| IPTorrents cookie-flow<br>component ( <code>iptorrents-cookie-flow.component.ts</code> )                                                                       | implemented | yes   | vitest (jackett specs)                                          | unit<br>(vitest) |       |

## 5e. Frontend infra

| Feature                            | Impl  | Wired | Tests            | Validation                   | Video                     |
|------------------------------------|-------|-------|------------------|------------------------------|---------------------------|
| Frontend<br>coverage<br>thresholds | infra | yes   | vitest.config.ts | enforced in<br>vitest config | N/A (no<br>UI —<br>build/ |

| Feature                                             | Impl | Wired | Tests | Validation | Video              |
|-----------------------------------------------------|------|-------|-------|------------|--------------------|
| (40%<br>lines/<br>branches/<br>funcs/<br>stmts, v8) |      |       |       |            | coverage<br>infra) |

## 6. BobaLink browser extension (extension/)

WXT + TypeScript Manifest-V3. Detects magnet/.torrent links and forwards to merge service :7187. Status detail: docs/browser\_extension/Status.md. Built zips at extension/.output/bobalink-1.0.0-{chrome,firefox}.zip.

### 6a. User-facing surfaces

| Feature                                                            | Impl        | Wired | Tests                              | Validation            | Video                                                                                                            |
|--------------------------------------------------------------------|-------------|-------|------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------|
| Popup — per-row Send button (src/popup/, entrypoints/popup.ts)     | implemented | yes   | ally tests/ally/popup.ally.test.ts | tested-green-in-suite | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b extension-pop verdict docs/qa recordings-20 boba-extension verdict.md |
| Popup — Send-All button                                            | implemented | yes   | ally popup.ally.test.ts            | tested-green-in-suite | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b extension-pop verdict docs/qa recordings-20 boba-extension verdict.md |
| Popup — Refresh button + status indicator (online/offline/warning) | implemented | yes   | ally popup.ally.test.ts            | tested-green-in-suite | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b extension-pop verdict docs/qa recordings-20 boba-extension verdict.md |
| Popup — torrent list (name + type + infohash prefix + sent badge)  | implemented | yes   | ally popup.ally.test.ts            | tested-green-in-suite | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b extension-pop verdict docs/qa recordings-20 boba-extension verdict.md |
| Popup — Open Options link                                          | implemented | yes   | ally popup.ally.test.ts            | tested-green-in-suite | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b extension-pop verdict docs/qa                                         |

| Feature                                                                                                             | Impl        | Wired | Tests                                                                           | Validation                    | Video                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------|-------------|-------|---------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Options page<br>(src/options/<br>options.ts)                                                                        | implemented | yes   | ally<br>options.ally.test.ts;<br>integration options-<br>background.test.ts     | tested-<br>green-in-<br>suite | recordings-2026-06-16 — boba-extension-verdict.md<br><b>VIDEO-CONF</b><br><b>2026-06-16</b> — boba-extension-options.mp4; docs/qa/recordings-2026-06-16 — boba-extension-verdict.md |
| Content script —<br>link/text<br>highlight overlay<br>(src/content/<br>highlight.ts,<br>entrypoints/<br>content.ts) | implemented | yes   | integration content-<br>background.test.ts;<br>security content-<br>xss.test.ts | tested-<br>green-in-<br>suite | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — boba-extension-scan/p detect.mp4; docs/qa/recordings-2026-06-16 — boba-extension-verdict.md                                                |
| Background<br>service worker<br>— message hub<br>(src/<br>background/<br>entrypoints/<br>background.ts)             | implemented | yes   | integration tests/<br>integration/*-<br>background.test.ts                      | tested-<br>green-in-<br>suite | N/A (no UI — test covered + exercised)<br>extension scan/p                                                                                                                          |
| Context-menu<br>entries (right-<br>click torrent<br>actions)                                                        | implemented | yes   | integration *-<br>background.test.ts                                            | tested-<br>green-in-<br>suite | N/A (no UI — test covered + exercised)<br>extension scan/p                                                                                                                          |
| Keyboard<br>shortcuts (scan-<br>now / highlight-<br>toggle / send-all)                                              | implemented | yes   | integration *-<br>background.test.ts                                            | tested-<br>green-in-<br>suite | N/A (no UI — test covered + exercised)<br>extension scan/p                                                                                                                          |
| Action badge<br>(detected count)<br>+ notifications                                                                 | implemented | yes   | integration *-<br>background.test.ts                                            | tested-<br>green-in-<br>suite | N/A (no UI — test covered + exercised)<br>extension scan/p                                                                                                                          |

## 6b. Scanners (src/scanner/)

| Feature                                                                | Impl        | Wired | Tests       | Validation                    | Video                                                      |
|------------------------------------------------------------------------|-------------|-------|-------------|-------------------------------|------------------------------------------------------------|
| Base scanner —<br>computeStableId()<br>lifecycle (scanner/<br>base.ts) | implemented | yes   | vitest unit | tested-<br>green-in-<br>suite | N/A (no UI — test-covered scanner; exercised by extension) |

| Feature                                                                            | Impl        | Wired | Tests                                                             | Validation            | Video                                                                                  |
|------------------------------------------------------------------------------------|-------------|-------|-------------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------|
| Link scanner — <a href> magnet/.torrent (scanner/link-scanner.ts)                  | implemented | yes   | security scanner-hostile-input.test.ts; perf scanner.perf.test.ts | tested-green-in-suite | scan-detect)<br>N/A (no UI — test-covered scanner; exercised by extension scan-detect) |
| Text scanner — bare magnet text (scanner/text-scanner.ts)                          | implemented | yes   | security scanner-hostile-input.test.ts                            | tested-green-in-suite | N/A (no UI — test-covered scanner; exercised by extension scan-detect)                 |
| Scanner orchestrator — dedup + mutation-observer re-scan (scanner/orchestrator.ts) | implemented | yes   | perf orchestrator-scaling.perf.test.ts                            | tested-green-in-suite | N/A (no UI — test-covered scanner; exercised by extension scan-detect)                 |
| Site DB — per-site selector match rules (scanner/site-db.ts)                       | implemented | yes   | vitest unit                                                       | tested-green-in-suite | N/A (no UI — test-covered scanner; exercised by extension scan-detect)                 |

## 6c. Parsers (src/parser/)

| Feature                            | Impl        | Wired | Tests                                                                   | Validation            | Video                      |
|------------------------------------|-------------|-------|-------------------------------------------------------------------------|-----------------------|----------------------------|
| Bencode parser (parser/bencode.ts) | implemented | yes   | security bencode-torrentfile-hostile.test.ts; perf parsers.perf.test.ts | tested-green-in-suite | N/A (no UI — test-covered) |



| Feature                                      | Impl        | Wired | Tests                                                                 | Validation            | Video                                                                                                                                                                                         |
|----------------------------------------------|-------------|-------|-----------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Magnet parser (parser/magnet.ts)             | implemented | yes   | security infohash-detection-hostile.test.ts; perf magnet.perf.test.ts | tested-green-in-suite | parser; exercised by extension scan-detect)<br>N/A (no UI — test-covered parser; exercised by extension scan-detect)<br>N/A (no UI — test-covered parser; exercised by extension scan-detect) |
| Torrent-file parser (parser/torrent-file.ts) | implemented | yes   | security bencode-torrentfile-hostile.test.ts                          | tested-green-in-suite | N/A (no UI — test-covered parser; exercised by extension scan-detect)                                                                                                                         |

## 6d. API client + offline (src/api/)

| Feature                                                                          | Impl        | Wired | Tests                                                               | Validation            | Video                                                                                                                                |
|----------------------------------------------------------------------------------|-------------|-------|---------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Boba API client — POST /api/v1/download, retry + rate-limit (api/boba-client.ts) | implemented | yes   | vitest unit; chaos tests/chaos/boba-client-resilience.chaos.test.ts | tested-green-in-suite | N/A (no UI — test-covered API client; exercised by extension Send)<br>N/A (no UI — test-covered health probe; exercised by extension |
| Health probe — probeHealth() (api/health.ts)                                     | implemented | yes   | vitest unit                                                         | tested-green-in-suite |                                                                                                                                      |

| Feature                                                             | Impl        | Wired | Tests                                 | Validation            | Video                                                                        |
|---------------------------------------------------------------------|-------------|-------|---------------------------------------|-----------------------|------------------------------------------------------------------------------|
| Offline send queue — IndexedDB enqueue + re-dispatch (api/queue.ts) | implemented | yes   | chaos tests/chaos/queue.chaos.test.ts | tested-green-in-suite | status indicator)<br>N/A (no UI — test-covered offline queue; chaos-covered) |

## 6e. Shared utilities + tabgroups (src/shared/, src/tabgroups/)

| Feature                                                                      | Impl        | Wired | Tests                                                       | Validation            | Video                                                                     |
|------------------------------------------------------------------------------|-------------|-------|-------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------|
| Crypto — AES-256-GCM token encrypt/decrypt (shared/crypto.ts)                | implemented | yes   | security crypto-tamper.test.ts, no-hardcoded-secret.test.ts | tested-green-in-suite | N/A (no UI — test-covered crypto; exercised by extension Send token path) |
| Storage — namespaced chrome.storage.local wrapper (shared/storage.ts)        | implemented | yes   | vitest unit                                                 | tested-green-in-suite | N/A (no UI — test-covered storage wrapper)                                |
| Constants — site selectors / rate-limit / retry config (shared/constants.ts) | implemented | yes   | covered via scanner tests                                   | tested-green-in-suite | N/A (no UI — test-covered constants)                                      |
| Errors — Network/Server/Storage error types (shared/errors.ts)               | implemented | yes   | vitest unit                                                 | tested-green-in-suite | N/A (no UI — test-covered error types)                                    |
| Events — TypedEventEmitter bus (shared/events.ts)                            | implemented | yes   | vitest unit                                                 | tested-green-in-suite | N/A (no UI — test-covered event bus)                                      |
|                                                                              | implemented | yes   | vitest unit                                                 |                       |                                                                           |

| Feature                                                       | Impl        | Wired | Tests                                            | Validation            | Video                                                                                     |
|---------------------------------------------------------------|-------------|-------|--------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------|
| Logger — namespaced, no-secret (shared/logger.ts)             |             |       |                                                  | tested-green-in-suite | N/A (no UI — test-covered logger)                                                         |
| Utils — debounce / TokenBucket rate limiter (shared/utils.ts) | implemented | yes   | vitest unit                                      | tested-green-in-suite | N/A (no UI — test-covered utils/rate-limiter)                                             |
| Tab groups — batch grouped-tab torrents (tabgroups/index.ts)  | implemented | yes   | vitest unit (tab-group Challenge per ext Status) | tested-green-in-suite | N/A (no UI of its own — test-covered tab-group batching; exercised by extension Send-All) |

## 6f. i18n locales (extension/src/public/\_locales/)

| Feature             | Impl        | Wired | Tests                                 | Validation            | Video                                                                                            |
|---------------------|-------------|-------|---------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------|
| Locale en (English) | implemented | yes   | i18n tests/i18n/locale-safety.test.ts | tested-green-in-suite | N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension popup/options) |
| Locale de (German)  | implemented | yes   | i18n locale-safety.test.ts            | tested-green-in-suite | N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension                |

| Feature                 | Impl        | Wired | Tests                          | Validation                    | Video                                                                                                                                                            |
|-------------------------|-------------|-------|--------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Locale es<br>(Spanish)  | implemented | yes   | i18n locale-<br>safety.test.ts | tested-<br>green-in-<br>suite | popup/<br>options)<br><br>N/A (no<br>UI of its<br>own —<br>i18n<br>locale<br>data;<br>test-<br>covered;<br>surfaced<br>inside<br>extension<br>popup/<br>options) |
| Locale fr<br>(French)   | implemented | yes   | i18n locale-<br>safety.test.ts | tested-<br>green-in-<br>suite | N/A (no<br>UI of its<br>own —<br>i18n<br>locale<br>data;<br>test-<br>covered;<br>surfaced<br>inside<br>extension<br>popup/<br>options)                           |
| Locale it<br>(Italian)  | implemented | yes   | i18n locale-<br>safety.test.ts | tested-<br>green-in-<br>suite | N/A (no<br>UI of its<br>own —<br>i18n<br>locale<br>data;<br>test-<br>covered;<br>surfaced<br>inside<br>extension<br>popup/<br>options)                           |
| Locale ja<br>(Japanese) | implemented | yes   | i18n locale-<br>safety.test.ts | tested-<br>green-in-<br>suite | N/A (no<br>UI of its<br>own —<br>i18n<br>locale<br>data;<br>test-<br>covered;<br>surfaced<br>inside                                                              |

| Feature                | Impl        | Wired | Tests                      | Validation            | Video                                                                                                                        |
|------------------------|-------------|-------|----------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| Locale pt (Portuguese) | implemented | yes   | i18n locale-safety.test.ts | tested-green-in-suite | extension popup/options)<br>N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension popup/options) |
| Locale ru (Russian)    | implemented | yes   | i18n locale-safety.test.ts | tested-green-in-suite | N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension popup/options)                             |

## 6g. Live round-trip

| Feature                                                                       | Impl        | Wired            | Tests                               | Validation                        | Video                                              |
|-------------------------------------------------------------------------------|-------------|------------------|-------------------------------------|-----------------------------------|----------------------------------------------------|
| Live download endpoint round-trip (tests/live/download-endpoint.live.test.ts) | implemented | yes (live-gated) | live download-endpoint.live.test.ts | partial — operator-gated live env | N/A (UI — live round-trip to operator-gated live e |

## 7. WebUI bridge (webui-bridge.py) — host process :7188

Bridges qBittorrent WebUI with private-tracker auth. NOT a container.

| Feature                                                              | Impl        | Wired | Tests                                                                                               | Validation            |
|----------------------------------------------------------------------|-------------|-------|-----------------------------------------------------------------------------------------------------|-----------------------|
| WebUI proxy + header rewrite (webui-bridge.py)                       | implemented | host  | integration test_webui_bridge.py, test_bridge_header_rewrite.py; stress test_bridge_stress_chaos.py | tested-green-in-suite |
| Private-tracker auth bridging (rutracker/kinozal/nnmclub/iptorrents) | implemented | host  | integration test_webui_bridge.py                                                                    | partial — live-gated  |
| Theme injection into bridged WebUI (plugins/theme_injector.py)       | implemented | host  | integration test_bridge_theme_injection.py                                                          | tested-green-in-suite |
| Bridge root liveness (/)                                             | implemented | host  | integration test_bridge_root_liveness.py                                                            | tested-green-in-suite |

## 8. Infrastructure / CLI / shell scripts

### 8a. boba-ctl CLI orchestrator (Go: cmd/boba-ctl/main.go + wrapper scripts/boba-ctl.sh)

| Feature                                             | Impl        | Wired | Tests                               | Validation            | Video                                                                            |
|-----------------------------------------------------|-------------|-------|-------------------------------------|-----------------------|----------------------------------------------------------------------------------|
| boba-ctl up — start stack (cmd/boba-ctl/main.go:32) | implemented | host  | go cmd/boba-ctl/projectroot_test.go | tested-green-in-suite | N/A (CLI — go-covered; exercise boba-ctl CLI den boba-cli-orchestrator-demo.mp4) |

| Feature                                                                                                                       | Impl        | Wired | Tests                                   | Validation                              | Video                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------|-------------|-------|-----------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| boba-ctl down<br>— stop stack<br>(main.go:34)                                                                                 | implemented | host  | go cmd/boba-ctl/<br>projectroot_test.go | tested-<br>green-in-<br>suite           | N/A (CLI — go-<br>covered; exercise<br>boba-ctl CLI dem<br>boba-cli-<br>orchestrator-<br>demo.mp4)                      |
| boba-ctl<br>status — service<br>status<br>(main.go:36)                                                                        | implemented | host  | go cmd/boba-ctl/<br>projectroot_test.go | tested-<br>green-in-<br>suite +<br>LIVE | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b<br>cli-orchestra<br>demo.mp4; verdi<br>qa/<br>recordings-20<br>boba-cli-verd |
| boba-ctl<br>health — health<br>probe<br>(main.go:38,<br>cmd/boba-ctl/<br>health_test.go)                                      | implemented | host  | go cmd/boba-ctl/<br>health_test.go      | tested-<br>green-in-<br>suite +<br>LIVE | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b<br>cli-orchestra<br>demo.mp4                                                 |
| boba-ctl list<br>— list services<br>(main.go:40)                                                                              | implemented | host  | go cmd/boba-ctl/<br>projectroot_test.go | tested-<br>green-in-<br>suite +<br>LIVE | <b>VIDEO-CONF</b><br><b>2026-06-16</b> — b<br>cli-orchestra<br>demo.mp4                                                 |
| boba-ctl.sh<br>wrapper —<br>compose-compat<br>shim (up/down/ps/<br>config/pull +<br>passthrough)<br>(scripts/boba-<br>ctl.sh) | implemented | host  | covered via boba-ctl go<br>tests        | tested-<br>green-in-<br>suite           | N/A (CLI shim —<br>covered; exercise<br>boba-ctl CLI dem                                                                |

## 8b. Lifecycle scripts (repo root)

| Feature                                                                        | Impl        | Wired | Tests                                                 | Validation                    |
|--------------------------------------------------------------------------------|-------------|-------|-------------------------------------------------------|-------------------------------|
| start.sh —<br>container bring-<br>up + Jaxxnet key<br>extraction/<br>injection | implemented | host  | covered via e2e<br>test_fixtures_bring_up_services.py | tested-<br>green-in-<br>suite |
| stop.sh — stop/<br>remove/purge                                                | implemented | host  | none dedicated                                        | not-<br>validated             |
| setup.sh —<br>one-time setup                                                   | implemented | host  | none dedicated                                        | not-<br>validated             |

| Feature                                                                                             | Impl        | Wired | Tests                           | Validation                    |
|-----------------------------------------------------------------------------------------------------|-------------|-------|---------------------------------|-------------------------------|
| start-proxy.sh<br>— proxy<br>container entry<br>helper                                              | implemented | host  | none dedicated                  | not-<br>validated             |
| install-<br>plugin.sh —<br>copy curated<br>plugins →<br>container engines                           | implemented | host  | covered indirectly (e2e search) | tested-<br>green-in-<br>suite |
| init-qbit-<br>password.sh —<br>qBit admin<br>password<br>bootstrap                                  | implemented | host  | none dedicated                  | not-<br>validated             |
| fix-qbit-<br>password.sh —<br>qBit admin<br>password repair                                         | implemented | host  | none dedicated                  | not-<br>validated             |
| setup-webui-<br>bridge-<br>service.sh +<br>webui-<br>bridge.service<br>— systemd unit<br>for bridge | implemented | host  | none dedicated                  | not-<br>validated             |

## 8c. Test / CI wrappers (repo root)

| Feature                                                   | Impl        | Wired | Tests                       | Validation                                                          | Video                                                                |
|-----------------------------------------------------------|-------------|-------|-----------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------|
| ci.sh — manual CI<br>(syntax+unit+integration+e2e+health) | implemented | host  | self<br>(is<br>the<br>gate) | tested-<br>green-in-<br>suite                                       | N/A<br>(host<br>CI/test<br>wrapper<br>— no<br>UI; is<br>the<br>gate) |
| run-all-tests.sh — full suite<br>(hardcoded podman)       | implemented | host  | self                        | partial —<br>fails on<br>docker-<br>only hosts<br>(known<br>gotcha) | N/A<br>(host<br>CI/test<br>wrapper<br>— no<br>UI; is<br>the<br>gate) |



| Feature                                                 | Impl        | Wired | Tests | Validation            | Video                                              |
|---------------------------------------------------------|-------------|-------|-------|-----------------------|----------------------------------------------------|
| <code>test.sh</code> — quick validation wrapper         | implemented | host  | self  | tested-green-in-suite | N/A<br>(host CI/test wrapper — no UI; is the gate) |
| <code>test-all.sh</code> — full validation wrapper      | implemented | host  | self  | tested-green-in-suite | N/A<br>(host CI/test wrapper — no UI; is the gate) |
| <code>test-full.sh</code> — extended validation wrapper | implemented | host  | self  | tested-green-in-suite | N/A<br>(host CI/test wrapper — no UI; is the gate) |

## 8d. scripts/ helpers

| Feature                                                                            | Impl        | Wired | Tests          | Validation            | Video                            |
|------------------------------------------------------------------------------------|-------------|-------|----------------|-----------------------|----------------------------------|
| <code>scripts/add-submodules.sh</code> — submodule bootstrap                       | implemented | host  | none dedicated | not-validated         | N/A<br>script UI; to cover opera |
| <code>scripts/audit-plugins.sh</code> — plugin discrepancy audit                   | implemented | host  | none dedicated | not-validated         | N/A<br>script UI; to cover opera |
| <code>scripts/build-releases.sh</code> — build extension/Go release artifacts      | implemented | host  | none dedicated | not-validated         | N/A<br>script UI; to cover opera |
| <code>scripts/codegraph_validate.sh</code> — CodeGraph index validation (§11.4.78) | implemented | host  | self           | tested-green-in-suite | N/A<br>script UI; to cover opera |
|                                                                                    | implemented | host  | self           |                       |                                  |

| Feature                                                              | Impl        | Wired | Tests                                                | Validation                            | Video                                     |
|----------------------------------------------------------------------|-------------|-------|------------------------------------------------------|---------------------------------------|-------------------------------------------|
| scripts/docs_chain.sh — docs_chain engine wrapper (§11.4.106)        |             |       |                                                      | tested-green-in-suite                 | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/ensure-macos-tunnel.sh — 0.0.0.0 SSH tunnel                  | implemented | host  | none dedicated (log qa-results/tunnel-keepalive.log) | not-validated; LAN threat-model RW-05 | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/tunnel-keepalive.sh — tunnel self-heal                       | implemented | host  | none dedicated                                       | not-validated                         | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/freeze-openapi.sh — freeze boba-jackett OpenAPI snapshot     | implemented | host  | contract tests/contract/openapi_test.go              | tested-green-in-suite                 | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/generate_markdown_exports.sh — md→html/pdf export (§11.4.65) | implemented | host  | self                                                 | tested-green-in-suite                 | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/helixqa.sh — HelixQA harness driver                          | implemented | host  | self                                                 | not-validated                         | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/opencode-helixqa.sh — OpenCode HelixQA driver                | implemented | host  | self                                                 | not-validated                         | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/install_git_hooks.sh — git hook installer (§11.4.75)         | implemented | host  | self                                                 | tested-green-in-suite                 | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/pre_build_verification.sh — pre-build gate sweep             | implemented | host  | self (is the gate)                                   | tested-green-in-suite                 | N/A<br>script<br>UI; to<br>cover<br>opera |
| scripts/pre_code_review.sh — code-review gate (§11.4.125)            | implemented | host  | self                                                 | tested-green-in-suite                 | N/A<br>script<br>UI; to                   |

| Feature                                                                     | Impl        | Wired | Tests          | Validation                    | Video                                                     |
|-----------------------------------------------------------------------------|-------------|-------|----------------|-------------------------------|-----------------------------------------------------------|
| scripts/<br>run_all_challenges.sh —<br>challenge bank runner                | implemented | host  | self           | not-<br>validated             | cover<br>opera<br>N/A<br>scrip<br>UI; t<br>cover<br>opera |
| scripts/run-tests.sh — test<br>runner helper                                | implemented | host  | self           | tested-<br>green-in-<br>suite | N/A<br>scrip<br>UI; t<br>cover<br>opera                   |
| scripts/scan.sh — repo scan<br>helper                                       | implemented | host  | none dedicated | not-<br>validated             | N/A<br>scrip<br>UI; t<br>cover<br>opera                   |
| scripts/track-build-<br>resources.sh — build resource<br>sampler (§11.4.24) | implemented | host  | none dedicated | not-<br>validated             | N/A<br>scrip<br>UI; t<br>cover<br>opera                   |

## 8e. Infra / compose / startup

| Feature                                                                                     | Impl        | Wired | Tests                                          | Validation                    | Video                      |
|---------------------------------------------------------------------------------------------|-------------|-------|------------------------------------------------|-------------------------------|----------------------------|
| docker-<br>compose.yml — 2-<br>container default +<br>Go profile + boba-<br>jackett service | infra       | yes   | e2e test_live_containers.py                    | tested-<br>green-in-<br>suite | L<br>n<br>h<br>t<br>C      |
| Jackett auto-<br>configuration at<br>startup (key extract<br>→ inject<br>JACKETT_API_KEY)   | implemented | yes   | integration<br>test_jackett_autoconfig_real.py | tested-<br>green-in-<br>suite | L<br>n<br>h<br>t<br>s<br>a |
| BOBA_MASTER_KEY<br>auto-generation<br>(first-boot) for<br>encrypted boba.db                 | implemented | yes   | go bootstrap/bootstrap_test.go                 | unit (go)                     | L<br>n<br>t<br>s           |

## Components NOT inventoried / discrepancies (honest gaps, §11.4.6)

- **CORRECTION vs Rev 2:** `boba-ctl.sh` DOES exist at `scripts/boba-ctl.sh` (Rev 2 said it did not). Its subcommands are now itemized in §8a. The underlying Go binary is `cmd/boba-ctl/main.go` (5 subcommands: `up/down/status/health/list`).
- **CORRECTION vs Rev 2:** the `install-plugin.sh` `PLUGINS` curated array has **44 entries**, not 12. ~21 have a matching `plugins/*.py` file; the rest (incl. `torrentproject`, `torrentscsv`) are curated names with no file present in this tree — itemized in §4c, not asserted as working.
- **Prometheus/OpenTelemetry metrics** — `tests/observability/test_metrics_exist.py` + `observability/` dir exist, but no `prometheus_client Counter()/Histogram()/Gauge()` definitions were found in `download-proxy/src`; metrics surface is via `GET /api/v1/stats` (counters maintained in-process), not a `/metrics` Prometheus endpoint. Stated as fact, not assumed.
- **Go DownloadHandler / ActiveDownloadsHandler** — flagged partial/stub in §2a (mock success / empty array, no qBit integration) per source reading. The Go profile is opt-in and NOT the shipped product.
- **Go scheduler** — CRUD-only, no driver loop (RW-10); schedules never fire. The Go enricher is missing (RW-11).
- **Video-recording confirmation** — PENDING for all rows EXCEPT the `boba-ctl` CLI subcommands (`status/health/list`) and the web-UI flows already filmed (`POST /api/v1/search journey` + `dashboard tab/button/theme/Jackett tour`). Remaining per-feature flows are an in-progress recording pass (§11.4.107/§11.4.143).